MODULE 5 NERVOUS TISSUE

Functional Organization and Nervous Tissue.

FUNCTIONAL ORGANIZATION AND NERVOUS TISSUE

Covers the organization of the nervous system and its divisions, the three functions, the structure of a neuron, the synapse, the classification of neurons, the neuroglia, myelination, and the collections of nervous tissue. The focus is on the structures and the job each one does.

BY THE END

1. Outline the organization of the nervous system and its divisions.
2. Identify the parts of a neuron and classify neurons by shape and by function.
3. Name the neuroglia and describe myelination and the collections of nervous tissue.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Nervous tissue

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

The Nervous System, an Overview

The nervous system gives the body the ability to feel, think, and act. It has two main divisions: the central nervous system and the peripheral nervous system.

THE TWO DIVISIONS AND THE STRUCTURES IN EACH

Structure	What it is
Nervous system	The body's fast control system; it lets us feel, think, and act
Central nervous system (CNS)	The brain and the spinal cord; it processes information and issues commands
Brain	The CNS organ in the cranial cavity, with about 85 billion neurons
Spinal cord	The CNS cord in the vertebral canal that links the brain to the body
Peripheral nervous system (PNS)	Every neural structure outside the brain and spinal cord
Nerves	Bundles of axons in the PNS; 12 pairs of cranial nerves and 31 pairs of spinal nerves
Ganglia	Small masses of neuron cell bodies located outside the CNS
Sensory receptors	Structures that monitor changes inside and outside the body

Divisions of the Peripheral Nervous System

The motor side of the peripheral nervous system is split by the kind of effector it controls. The somatic system runs skeletal muscle; the autonomic system runs the viscera; the enteric system runs the gut.

THE MOTOR DIVISIONS OF THE PNS

Structure	What it is
Somatic nervous system (SNS)	The voluntary division; its motor neurons carry commands from the CNS to skeletal muscle
Autonomic nervous system (ANS)	The involuntary division; its motor neurons carry commands to smooth muscle, cardiac muscle, and glands
Sympathetic division	The ANS division for fight or flight, supporting exercise and emergencies
Parasympathetic division	The ANS division for rest and digest, supporting routine activities
Enteric nervous system (ENS)	The brain of the gut, roughly 100 million neurons in the wall of the GI tract that can operate on their own

The Three Functions of the Nervous System

Whatever the task, the nervous system handles it in the same three steps. A stimulus comes in, the information is processed, and a response goes out.

1. **Sensory function** sensory receptors detect a change inside or outside the body and send the input toward the CNS
2. **Integrative function** the CNS analyzes the sensory input, stores it, and decides on a response
3. **Motor function** the CNS sends commands out to the effectors, the muscles and glands, which carry out the response

The Structure of a Neuron

A neuron is the signaling cell of the nervous system. It is excitable, it can carry an impulse, and it cannot divide. Every neuron has three basic parts.

THE PARTS OF A NEURON

Structure	What it is
Neuron	The excitable cell that receives, conducts, and transmits nerve impulses; neurons cannot divide
Cell body	The soma or perikaryon; it holds the nucleus and cytoplasm and is the metabolic center of the neuron
Dendrites	The branched, tree-like processes that receive incoming signals
Axon	The single long process that carries the nerve impulse away toward other cells
Nissl bodies	Clusters of rough ER and ribosomes in the cell body; the site of protein synthesis
Neurofibrils	Cytoskeletal fibers that give the neuron its shape and support
Microtubules	Cytoskeletal tubes that transport materials along the neuron
Axon hillock	The cone-shaped region where the cell body joins the axon

Structure	What it is
Axoplasm and axolemma	The cytoplasm of the axon and the plasma membrane of the axon
Axon terminals	The fine branches at the far end of the axon that contact the target cell
Lipofuscin	A harmless pigment that builds up in neurons with age

The Synapse

A synapse is the junction where a neuron passes its signal to another neuron, a muscle, or a gland. At a chemical synapse the signal crosses as a chemical message.

THE SYNAPSE AND ITS PARTS

Structure	What it is
Synapse	The site where a neuron communicates with another neuron, a muscle cell, or a gland
Axodendritic synapse	An axon ending on a dendrite
Axosomatic synapse	An axon ending on a cell body
Axoaxonic synapse	An axon ending on another axon
Chemical synapse	A synapse with no direct contact; the signal crosses as neurotransmitter molecules
Synaptic cleft	The narrow gap between the two cells that the neurotransmitter diffuses across
Synaptic end bulbs	The swellings at the axon terminal that hold the signal-sending machinery
Synaptic vesicles	The tiny sacs inside the end bulbs that store neurotransmitter
Neurotransmitters	The chemical messengers that carry the signal across the cleft to the next cell

Classification of Neurons

Neurons are classified two ways: by their shape, which is the structural classification, and by the direction they carry signals, which is the functional classification.

Structural classification, by shape

NEURONS NAMED FOR THE NUMBER OF PROCESSES

Structure	What it is
Multipolar neuron	Many dendrites and one axon; the most common type, found in the brain and cord and as motor neurons
Bipolar neuron	One dendrite and one axon; found in the retina, the inner ear, and the olfactory area
Unipolar neuron	A single fused process; the typical shape of a sensory neuron
Purkinje cells	A distinctive multipolar neuron found in the cerebellum
Pyramidal cells	A distinctive multipolar neuron found in the cerebral cortex

Functional classification, by direction

NEURONS NAMED FOR THE DIRECTION THEY CARRY A SIGNAL

Structure	What it is
Sensory (afferent) neurons	Carry sensory input toward the CNS; most are unipolar
Motor (efferent) neurons	Carry motor commands away from the CNS; these are multipolar
Interneurons	Also called association neurons; they connect sensory and motor neurons within the CNS and are multipolar

Neuroglia

Neuroglia, or glial cells, are the support cells of nervous tissue. They are smaller and far more numerous than neurons, and unlike neurons they can still divide.

WHAT THE GLIAL CELLS ARE

Structure	What it is
Neuroglia	The support cells of nervous tissue; they nourish, support, and protect neurons and make up about half the volume of the CNS

Glia of the central nervous system

THE FOUR GLIA OF THE CNS AND WHAT EACH ONE DOES

Glial cell	Role
Astrocytes	Star-shaped cells that give structural support, help form the blood-brain barrier, guide neuron growth, and maintain the chemical environment
Oligodendrocytes	Cells that form the myelin sheaths of the CNS
Microglia	Small phagocytic cells that remove debris and microbes
Ependymal cells	Cells that line the ventricles and central canal and help produce and circulate cerebrospinal fluid

Glia of the peripheral nervous system

THE TWO GLIA OF THE PNS AND WHAT EACH ONE DOES

Glial cell	Role
Schwann cells	Cells that form the myelin sheath around PNS axons and aid axon regeneration
Satellite cells	Cells that surround and support the cell bodies of PNS neurons in ganglia

Myelination

Myelin is a fatty wrapping around an axon. It insulates the axon and makes the nerve impulse travel much faster.

THE MYELIN SHEATH AND WHAT IT DOES

Structure	What it is
Myelin sheath	A multilayered covering of lipid and protein wrapped around an axon
Function of myelin	It insulates the axon and speeds up the nerve impulse
Nodes of Ranvier	The gaps in the myelin sheath, spaced along the axon
Saltatory conduction	The fast conduction in which the impulse jumps from node to node

The cell that builds the myelin differs between the PNS and the CNS, and that difference decides whether a damaged axon can regrow. Compare them.

PNS AND CNS MYELINATION COMPARED

Feature	PNS myelination	CNS myelination
Cell that makes the myelin	A Schwann cell, one per axon segment	An oligodendrocyte, one cell for up to 15 axons
Neurolemma	Present; it is the outer layer of the Schwann cell	Absent
Axon regeneration	Possible; the neurolemma guides the axon as it regrows	Very limited, because there is no neurolemma

Myelination increases from birth to maturity, so an infant's nerve conduction is slower than an adult's.

KEY POINT

One Schwann cell myelinates one segment of one axon and leaves a neurolemma behind. One oligodendrocyte reaches up to 15 axons and leaves none. That single difference is why a cut nerve in the arm can regrow and a cut tract in the cord usually cannot.

Collections of Nervous Tissue

The same kind of structure has a different name depending on whether it sits in the CNS or the PNS. Learn these as pairs.

THE SAME STRUCTURE UNDER TWO NAMES

Structure	In the CNS	In the PNS
A cluster of cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve

WHITE MATTER AND GRAY MATTER, AND WHERE EACH ONE SITS

Structure	What it is
White matter	Nervous tissue made of myelinated axons
Gray matter	Nervous tissue made of cell bodies, dendrites, unmyelinated axons, and neuroglia
In the spinal cord	White matter is on the outside, gray matter is on the inside
In the brain	Gray matter is on the outside, white matter is on the inside

KEY POINT

Gray is where the cell bodies are, white is where the myelin is. The two swap positions between the cord and the brain, so name the organ before you name the layer.

WHEN THE SUPPORT CELLS FAIL

Neurons cannot divide, so a tumor of nervous tissue almost never comes from neurons. It comes from the **glia**, which can still divide. A **glioma** is a tumor of glial cells, and it is the most common type of brain tumor.

Myelin is as important as the axon it wraps. In **multiple sclerosis**, the immune system strips myelin from CNS axons and signals slow or fail. The presence of the **neurolemma** on Schwann cells, and its absence on oligodendrocytes, is why a cut nerve in the arm can regrow but a cut tract in the cord usually cannot.

The **cell body** is the neuron's life-support. Because the axon depends on the cell body for proteins, an injury that separates the two kills the axon beyond the cut. The location of a nerve injury, near or far from the cell body, helps predict whether the neuron survives.